# Finding the Eigenvalues and Eigenfunctions of Integral Operators


Seunghyun Lee [1]

[1]Imperial College London


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## Motivation: The Eigenvalue / Eigenfunction Problem

Consider some (linear) integral operator $\mathcal{K} : L^2(a,b) \to L^2(a,b)$ such that

$$\mathcal{K}f(x) = \int_a^b K(x,y)f(y)\,dy$$

where $a, b$ are real numbers and $K(x,y)$ is a real function defined over $(a,b)$, called the **kernel**.

Finding the eigenvalues and eigenfunctions of operator $\mathcal{K}$ would involve solving the integral equation

$$\mathcal{K}f(x) = \int_a^b K(x,y)f(y)\,dy = \lambda f(x)$$

also known as the **homogeneous Fredholm equation of the 2nd kind**. We shall first look into Fredholm's theory, which provides a method of solving such equations for any kernel $K(x,y)$ that is bounded and integrable. Afterwards, we will discuss what other types of kernels can be studied for further generalisation.

## Separable Kernels

We begin by looking at the case where the given kernel $K(x,y)$ is **separable**, i.e. $K(x,y)$ is of the form

$$K(x,y) = \sum_{k=1}^{n} a_k(x)b_k(y)$$

where $\{a_k\}$ and $\{b_k\}$ are both linearly independent. Using that $K(x,y)$ is separable and $\{a_k\}$ is linearly independent, we obtain the following result.

### Theorem 1: Solution for Separable Kernels

Suppose $K(x,y)$ is separable. Then:

- $\mathcal{K}$ has eigenvalue $\lambda = 0$ with eigenfunction $f$ if and only if
$$\int_a^b f(x)b_k(x)\,dx = 0, \qquad k = 1, \cdots, n.$$
- $\mathcal{K}$ has eigenvalue $\lambda \neq 0$ if and only if
$$D(\mu) = \det\begin{pmatrix} 1-\mu a_{11} & -\mu a_{12} & \cdots & -\mu a_{1n} \\ -\mu a_{21} & 1-\mu a_{22} & \cdots & -\mu a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -\mu a_{n1} & -\mu a_{n2} & \cdots & 1-\mu a_{nn} \end{pmatrix} = 0$$
where $\mu = 1/\lambda$ and
$$a_{kl} = \int_a^b b_k(x)a_l(x)\,dx$$
and corresponding eigenfunction $f$ is of the form $f(x) = \mu\sum_{k=1}^n c_k a_k(x)$ where $c_1, \cdots, c_n$ satisfy
$$c_k - \mu\sum_{l=1}^{n} a_{kl}c_l = 0, \qquad k = 1, \cdots, n.$$

Also, the equation $f(x) = \mu\int_a^b K(x,y)f(y)\,dy$ has a unique solution $f(x) = 0$ if $D(\mu) \neq 0$.

### Example: $K(x,y)$ that is separable

If $K(x,y) = \sin(x+y)$ for $0 \le x, y \le \pi$, then we can write $K(x,y) = \sum_{k=1}^2 a_k(x)b_k(y)$ where

$$a_1(x) = \sin(x), \quad a_2(x) = \cos(x), \quad b_1(y) = \cos(y), \quad b_2(y) = \sin(y)$$

Now we find the eigenvalues and eigenfunctions:

- $\mathcal{K}$ has eigenvalue $\lambda = 0$ if and only if there exists function $f$ such that
$$\int_0^\pi f(x)\cos(x)\,dx = \int_0^\pi f(x)\sin(x)\,dx = 0$$
- Using Theorem 1, $D(\mu) = 1 - \pi^2\mu^2/4$ and $\lambda = \pm\pi/2$. For $\lambda = \pi/2$, its corresponding eigenfunctions are
$$f(x) = c(\sin(x) + \cos(x)), \qquad c \in \mathbb{R}$$
and for $\lambda = -\pi/2$, its corresponding eigenfunctions are
$$f(x) = c'(\sin(x) - \cos(x)), \qquad c' \in \mathbb{R}$$

## Fredholm's Method of Solution

Now we assume that $K(x,y)$ is any function that is bounded and integrable. Consider the equation

$$f(x) = \mu\int_a^b K(x,y)f(y)\,dy$$

The idea is to divide the interval $(a,b)$ into $n$ equal subintervals: let

$$x_1 = y_1 = a, \qquad x_2 = y_2 = a+h, \qquad \cdots, \qquad x_n = y_n = a+(n-1)h$$

where $h = (b-a)/n$. This leads to an 'approximate' system of linear equations

$$f_i - \mu h\sum_{j=1}^{n} K_{ij}f_j = 0, \qquad i = 1, \cdots, n,$$

where $f_i = f(x_i)$ and $K_{ij} = K(x_i, x_j)$. Next, find the associated determinant of this system:

$$D_n(\mu) = \det\left(\delta_{ij} - \mu hK_{ij}\right)_{1\le i,j\le n}$$

where $\delta_{ij}$ is the Kronecker delta. One way of finding the 'approximate' eigenvalues is by solving $D_n(\mu) = 0$; with smaller error bounds in general, as $n$ increases. [3]

### Example: Finding Approximate Eigenvalues

We will find the approximate eigenvalue of $K(x,y) = \sin(x+y)$ by solving $D_n(\mu) = 0$ for $n = 5$:

- let $x_i = (i-1)\pi/5$ where $i = 1, \cdots, 5$;
- find the matrix $A = (hK_{ij})_{1\le i,j\le 5}$ and calculate its characteristic polynomial;
- using Mathematica to solve the characteristic equation, we get the rounded values
$$\lambda_1 \approx 1.571, \quad \lambda_2 \approx -1.571, \quad \lambda_3, \lambda_4, \lambda_5 \approx 0,$$
which is a very accurate approximation of the eigenvalues found in the previous example.

As $n$ approaches infinity,

$$D_n(\mu) \to D(\mu) = 1 + \sum_{m=1}^{\infty}\frac{(-\mu)^m}{m!}\int_a^b\cdots\int_a^b K\begin{pmatrix} x_1 & x_2 & \cdots & x_m \\ x_1 & x_2 & \cdots & x_m\end{pmatrix}dx_1\cdots dx_m$$

and the series on the right hand side converges for all values of $\mu$, [3] where

$$K\begin{pmatrix} x_1 & x_2 & \cdots & x_n \\ y_1 & y_2 & \cdots & y_n\end{pmatrix} = \det\left(K(x_i, y_j)\right)_{1\le i,j\le n}$$

We call $D(\mu)$ the **resolvent determinant** of our integral equation, and $D(\mu)$ turns out to play the same role as the $D(\mu)$ we found for separate kernels did.

### Theorem 2: Fredholm Alternative for Homogeneous Equations

For a fixed value $\mu$, the homogeneous equation $f(x) = \mu\int_a^b K(x,y)f(y)\,dy$ has either:

- unique solution $f(x) = 0$ if $D(\mu) \neq 0$;
- or $1 \le r \le n$ linearly independent solutions if $D(\mu) = 0$.

Therefore, solving $D(\mu) = 0$ gives us all nonzero eigenvalues of $\mathcal{K}$. Now to finding the eigenfunctions: consider some value $\mu_0$ that is a zero of $D(\mu)$ with algebraic multiplicity $k$ (i.e., $D(\mu_0) = D'(\mu_0) = \cdots = D^{(k-1)}(\mu_0) = 0$ and $D^{(k)}(\mu_0) \neq 0$). Define the **$n$th Fredholm minor** as

$$D_n\left(\begin{matrix} x_1 & x_2 & \cdots & x_n \\ y_1 & y_2 & \cdots & y_n\end{matrix}\middle|\,\mu\right) = K\begin{pmatrix} x_1 & x_2 & \cdots & x_n \\ y_1 & y_2 & \cdots & y_n\end{pmatrix} + \sum_{m=1}^{\infty}\frac{(-\mu)^m}{m!}\int_a^b\cdots\int_a^b K\begin{pmatrix} x_1 & \cdots & x_n & t_1 & \cdots & t_m \\ y_1 & \cdots & y_n & t_1 & \cdots & t_m\end{pmatrix}dt_1\cdots dt_m$$

Expand the determinant integrand along the $i$th row and $j$th column. Then, use that the $k$th Fredholm minor is not identically zero to prove the following result.

### Theorem 3: Solution of Homogeneous Fredholm Equation

If $\mu_0$ is a zero of $D(\mu)$ with algebraic multiplicity $k$, then the homogeneous equation $f(x) = \mu_0\int_a^b K(x,y)f(y)\,dy$ has $r$ $(1 \le r \le k)$ linearly independent solutions of the form

$$f_i(x) = D_r\left(\begin{matrix} x_1 & \cdots & x_{i-1} & x & x_{i+1} & \cdots & x_r \\ y_1 & \cdots & y_{i-1} & y_i & y_{i+1} & \cdots & x_r\end{matrix}\middle|\,\mu\right), \qquad i = 1, \cdots, r.$$

### Example: Using Fredholm's Method

Again consider $K(x,y) = \sin(x+y)$ over $0 \le x, y \le \pi$. Using the formulas for the resolvent determinant $D(\mu)$ and the 1st Fredholm minor $D_1(x,y;\mu)$, we get:

- $D(\mu) = 1 - \dfrac{\mu^2\pi^2}{2!\ 2} = 1 - \pi^2\mu^2/4$
- $D_1(x,y;\mu) = \sin(x+y) + \mu\dfrac{\pi}{2}\cos(x-y)$

Therefore, the nonzero eigenvalues are $\lambda = \pm\pi/2$ and its corresponding eigenfunction is:

- for $\lambda = \pi/2$, $f(x) = \sin(x+y) + \cos(x-y)$ where $y$ is any value between 0 and $\pi$;
- for $\lambda = -\pi/2$, $f(x) = \sin(x+y) - \cos(x-y)$ where $y$ is any value between 0 and $\pi$.

## Bounded Linear Operators

- Real function $f$ defined on $(a,b) \subseteq \mathbb{R}$ is called a **$L^2$-function** (or **square-integrable**) over $(a,b)$ if
$$\int_a^b |f(x)|^2\,dx < \infty$$
- Operator $\mathcal{K}$ is **bounded** if there exists $c \in \mathbb{R}$ such that $\|\mathcal{K}f\| \le c\|f\|$ for all $f \in L^2(a,b)$. The **operator norm** of $\mathcal{K}$ is defined as
$$\|\mathcal{K}\| = \sup_{\|f\|\neq 0}\frac{\|\mathcal{K}f\|}{\|f\|}$$
- If $K(x,y)$ is an $L^2$-function over $(a,b)\times(a,b)$, i.e.
$$\|K\|^2 = \int_a^b\int_a^b |K(x,y)|^2\,dy\,dx < \infty$$
then $\mathcal{K}$ is a bounded operator and $|\lambda| \le \|\mathcal{K}\| \le \|K\|$.

Hence, all kernels we have dealt with so far are kernels of bounded operators. But the kernel does not necessarily have to be a $L^2$-function for its operator to be bounded.

### Example: Bounded Operator with Non-$L^2$ Kernel

Suppose $\mathcal{K} : L^2(0,\infty) \to L^2(0,\infty)$ such that

$$\mathcal{K}f(x) = \int_0^\infty \frac{1}{x+y}f(y)\,dy$$

Notice that

$$\int_0^\infty\int_0^\infty\left(\frac{1}{x+y}\right)^2dy\,dx \quad \text{diverges,}$$

implying that $K(x,y) = 1/(x+y)$ is not a $L^2$-function over $(0,\infty)\times(0,\infty)$. However,

$$\begin{aligned}(\mathcal{K}f, f) &= \int_0^\infty\int_0^\infty\frac{f(x)f(y)}{x+y}\,dy\,dx = \int_0^\infty\int_0^\infty\frac{f(x)f(tx)}{1+t}\,dt\,dx \\ &\le \|f\|^2\int_0^\infty\frac{1}{\sqrt{t}(1+t)}\,dt = \|f\|^2\int_0^\infty\frac{2}{1+u^2}\,du = \pi\|f\|^2\end{aligned}$$

where $t = y/x$, $u = \sqrt{t}$. Cauchy-Schwarz inequaliy is used for the second line.

Therefore, $\|\mathcal{K}\| \le \pi$; $\mathcal{K}$ is bounded.

Further study on functions like $K(x,y) = 1/(x+y)$ above $(0,\infty)$ and solving singular integral equations would lead to an even more general understanding of the eigenvalue / eigenfunction problem.

## References


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